

BALANCING ENERGY, ECONOMY, AND ENVIRONMENT: MULTI-OBJECTIVE RETROFIT FOR VÄSTRA BJÄRE PASTORAT

Himadri GULALIA, Kamal SINHA, and Vinay D JAYASELAN

¹Divisions of Energy and Building Design, Lund University, Sweden

²Division of Building Services, Lund University, Sweden

himadri.gulalia.2075@student.lu.se, kamal.sinha.5515@student.lu.se, vinay.dilip_j.6534@student.lu.se

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SUMMARY

This project examines energy retrofit strategies for Västra Bjäre Pastorat, a community centre and parish office located in Västra Karup, Skåne, Sweden. Given the proposed partial conversion of the building for residential use, the renovation offers an opportunity to enhance overall energy performance.

A key aspect of the renovation strategy is evaluating integration of renewable energy sources to offset the building's energy demand. Other energy-saving measures, including better insulation and new windows, are also assessed. The evaluation process was conducted in three core areas: energy performance, economic feasibility, and lifecycle environmental impact.

The goal is to develop a renovation strategy that balances overall energy efficiency improvements, cost expenditure, and carbon emission. The result shows that there is not one set of parameters that balances all three output performance criteria, but rather a set of options with respect to different building functions.

INTRODUCTION

1.1. Background

In recent years, there has been an increasing interest in energy renovations due to rising energy prices and global carbon dioxide (CO₂) emissions, finite non-renewable resources, and the need for a sustainable economy (Mata, Sasic Kalagasidis and Johnsson, 2014). In the context of Sweden, the growing demand for large-scale building renovations presents a unique opportunity to enhance energy efficiency, support national energy and climate targets, and provide modern, comfortable living spaces (Palm and Reindl, 2016).

This report presents an integrated analysis of potential energy renovation strategies for Västra Bjäre Pastorat, a public building, undertaken with consideration of energy, economic, and environmental feasibility. The aim is to identify an optimised and sustainable renovation design strategy with the primary objectives as follows:

- a. Implementation of Energy-efficient measures.
- b. Integration of Renewable Energy in terms of Photovoltaic Cells (P.V.).

- c. To perform Life Cycle Cost (L.C.C.) and Life Cycle Assessment (L.C.A.) analysis to determine optimal design solutions that minimize both economic and environmental impacts.
- d. To investigate the renovation of the part of the existing building into residential apartments.

1.2 Project description and overall purpose

The building serves multiple functions and comprises the following areas:

- a. First floor: A primary hall for parish activities, administrative offices, a dining hall with an adjacent kitchen, and two entrance foyers. One foyer provides access to washroom facilities and contains staircases to the upper and lower levels, while the other to the main hall.
- b. Second floor: Additional office space and a designated leisure area.
- c. Basement: A hobby room, a washroom, and a corridor providing access to a storage room and a mechanical/pump room.

METHODOLOGIES

This project investigates energy renovation strategies for a public-building through a four-phase methodology.

- Phase 1 establishes a performance baseline through site analysis and dynamic energy simulations.
- Phase 2 systematically explores renovation measures using parametric modelling to test passive strategies (insulation and windows), active strategies (photovoltaic integration), and behavioural scenarios (office-to-residential conversion).
- Phase 3 evaluates environmental and economic performance using Life Cycle Assessment and Life Cycle Costing over a 50-year period.
- Phase 4 integrates results to identify optimal renovation pathways balancing energy performance, environmental impact, and cost-effectiveness.

2.1 Building Analysis and Baseline Establishment (Phase 1):

The first phase involved analysis of the existing building to establish a performance baseline.

Data Collection and Analysis:

A site analysis was carried out to document the building's orientation, surrounding context, and shading obstructions. Existing architectural drawings, construction documents, operational schedules, and energy bills were also reviewed.

The building is a two-storey structure with a basement and an attic, providing a heated floor area of approximately 576.2 m². The analysed data indicates an Energy Use Intensity (EUI) of 57.1 kWh/m²·year. Heating is supplied entirely by electricity, with no district-heating connection. The building has no central ventilation system and instead uses individual exhaust fans. Occupancy varied throughout the year without a fixed schedule. Office spaces were assumed to be occupied for three hours on weekdays, with one-hour lunch break, while shared public areas showed fluctuating occupancy depending on the day, time, and user activity.

Digital Modelling:

Multiple computational tools and simulation programmes were used to analyse the performance of the building and improve it further. For the first phase, Rhinoceros 8 (Associates, 2025) was

used to create the base-case model to match real-life conditions. The Ängelholm-Helsingborg weather data was used for the climate analysis because it is the nearest available station to Västra Karup. (Google Maps, 2025).

Baseline Performance Simulation:

Grasshopper, along with plugins such as ClimateStudio 3.9 and other supportive tools, was used to simulate the building's current energy use and indoor thermal comfort. This created the benchmark against which all renovation measures were evaluated. Based on the simulation results, the Energy Use Intensity (EUI) was found to be 54.50 kWh/m²·year, which closely matches the reference data. The input parameters for the base case, such as envelope material specifications and energy loads are mentioned below (see Table 1):

Table 1: Common inputs of the base case and improved case.

Input Parameters	Existing
Equipment Loads & Lighting Loads /(W/m ²)	12
Heating Setpoint /°C	21.0
Cooling Setpoint /°C	30.0
U-Value of the Roof / (W/ (m ² .K))	0.15
U-Value of Internal Walls / (W/ (m ² .K))	4.65
U-Value of the Ground Slab / (W/ (m ² .K))	3.45
U-Value of the Basement wall / (W/ (m ² .K))	3.12

Although the technical documentation specifies a heat pump coefficient of performance (COP) of 4.72, a reduced value of 3 was applied in the digital model to better reflect real-world performance. Cooling, natural ventilation, and humidity control were disabled for this analysis.

Infiltration:

Carbon dioxide (CO₂) decay curves were generated from the provided hourly data. Infiltration rates were determined by analyzing carbon dioxide (CO₂) decay curves measured for each zone. The average value of each logarithmic curve was used to calculate air change rates (ACH) for both the baseline and improved scenarios. The resulting ACH values were 0.35 per hour for the Main Hall, Dining Hall, and Kitchen, 0.37 per hour for the first-floor offices, and 0.45 per hour for the basement, second-floor offices, attic, and service areas.

Thermal Zones:

The energy-use assessment was carried out by dividing the building into occupied, unoccupied, and semi-occupied thermal zones to improve the accuracy of the calculations. Areas with similar envelope construction and thermal conditions were grouped into the same thermal zones. Based on these criteria, a total of 10 zones were identified. (see Table 2)

Table 2: Thermal zones

Thermal Zones	Area Considered
Occupied	Main Hall, Dining Hall, and Offices/Apartments (Depending on the function). (see Fig. 2 & Fig. 3)
Unoccupied	Entry foyer, Lobby, circulation areas, kitchen, washroom/bathroom (depending on the function), and Attic. (see Fig. 1, Fig. 2 & Fig. 3)
Semi-Occupied	Basement hobby room. (see Fig. 1)

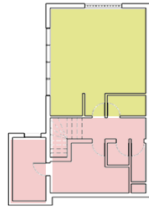


Fig. 1: Basement- Thermal Zones

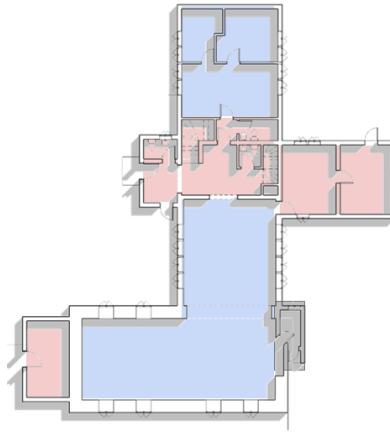


Fig. 2: First Floor- Thermal Zones

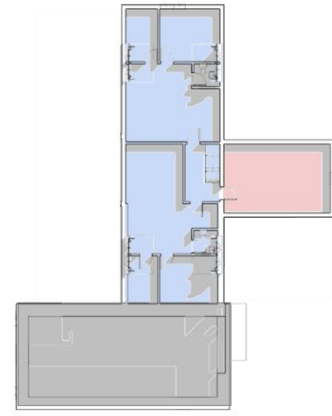


Fig. 3: Second Floor- Thermal Zones

2.2 Parametric Design and Simulation of Renovation Measures (Phase 2):

The investigation was carried out through parametric modelling and simulation, allowing the exploration of multiple design variables and their combinations.

Development of Parametric Workflow:

A parametric workflow was established in Grasshopper to systematically test combinations of renovation measures. Key parameters were grouped into Passive, Active, and Behavioural strategies as follows:

Passive strategies:

Passive strategies focus on minimizing the building's energy demand through architectural design and building envelope improvements, without relying on mechanical systems. These strategies use natural elements and materials to help regulate indoor temperature, humidity, and comfort. (Sarir and Sharifzadeh, 2024)

a. Insulation:

This study examines different insulation materials with varying thickness for the building envelop to reduce energy losses. Two insulation materials are considered: glass mineral wool (G.M.W.), a widely used synthetic option, and hemp fibre, a biogenic alternative. (see Table 3)

Table 3: Insulation Characteristics (Per 100 mm thickness)

Parameter	Mineral Wool Insulation (Knauf Glass)	Hemp Fiber Insulation (Ekolution)
Thermal Conductivity $/(W/(m \cdot K))$	0.03	0.04
Density $/(kg/m^3)$	30.70	30.00
Specific heat $/(J/(kg \cdot K))$	800	2300

For practicality, insulation is applied only in the office and apartment areas. However, the dining hall walls are also included in the analysis because they lie directly beneath the second-floor apartments. The insulation thickness varies from 100 mm to 300 mm, in 100 mm increments, and is applied to either the interior or exterior side to assess its impact on energy performance. Across all passive measures, a ventilated air gap was incorporated within the wall systems. This was done to address the moisture related issues while improving indoor air quality.

b. Window:

This study examines the replacement or improvement of window glazing and frames to enhance thermal comfort. For feasibility, only the windows of the offices/apartments, and the dining hall have been included in the analysis. Three window types are evaluated: the existing window as baseline, an upgraded double-glazed window with improved U-value, and a triple-glazed window. (see Table 4)

Table 4: Window Comparison

Parameter	Base Case Window	Double-Glazed Window (Window 1)	Triple-Glazed Window (Window 2)
Whole Window U-value $/(W/(m^2K))$	1.30	1.22	0.66
Solar Heat Gain Coefficient (between 0-1)	0.60	0.68	0.29
Visible Light Transmittance (T_{vis}) / %	13	27	16

Active Strategies:

Active strategies involve on-site renewable energy generation to balance or exceed annual energy consumption. These systems convert natural resources such as wind or solar energy into electricity to reduce or eliminate dependence on the grid. (Sarir and Sharifzadeh, 2024)

Photovoltaic (PV) Integration:

The PV integration strategy was developed by analysing the optimally oriented south-facing roof surfaces and dividing them into three installation scenarios based on peak power potential. Polycrystalline silicon (poly-Si) photovoltaic (PV) modules were selected for the building. For consistency, the PV panel dimensions were assumed to be 1950 mm $\times$ 1000 mm. (see Table 5)

Table 5: PV installation configurations

Srl.No	Location	Area $/m^2$	Efficiency $/\%$	Irradiance $/(W/m^2)$	Power $/kWp$	Number of Panels
1.	Above Attic	28	20%	1000	5.8	7
2.	Above Main Hall	52	20%	1000	11.0	13
3.	Combination 1. & 2.	80	20%	1000	16.9	20

The economic evaluation of the PV system is based on hourly electricity price variations for both purchasing electricity from the grid and selling surplus electricity back to it. Since Sweden operates under a dynamic pricing model, spot electricity prices fluctuate throughout the day in response to changes in demand, production levels, and market conditions. Incorporating hourly price data allows for a more accurate assessment of the PV system's financial performance.

Behavioural:

When the function of a building is altered, such as converting an office space into a residential apartment, occupancy patterns, people density, and usage profiles change, directly influencing the energy demand and indoor environmental conditions (Sarir and Sharifzadeh, 2024). For this scenario, the existing office spaces are proposed as residential apartments, covering a total area of 208.5 m². Of which, 51.5 m² is located on the first floor, with the remaining area situated on the second floor. The basement remains unchanged (see Fig. 4). The first floor contains a double-bedroom unit (see Fig. 5), while the second floor accommodates two one-bedroom units (see Fig. 6). The occupancy profiles for both apartments were taken from Climate Studio's default residential occupancy settings. The apartment retains the existing ventilation system, where the ventilation schedule was adopted from the Climate Studio default library.

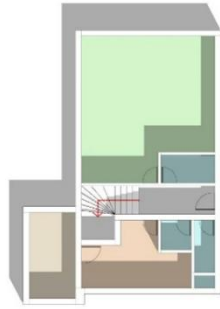


Fig. 4: Basement Floor Schematic Layout



Fig. 5: First Floor Proposed Schematic Layout

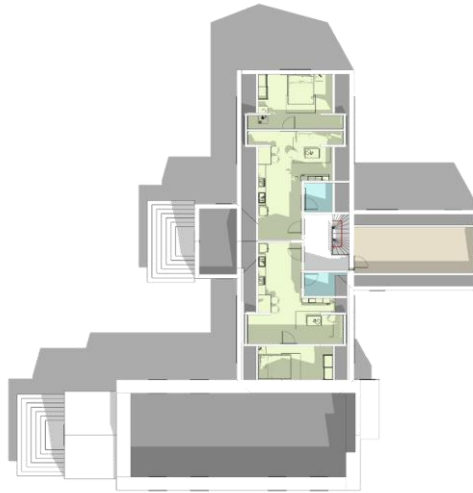


Fig. 6: Second Floor Proposed Schematic Layout

2.3 Sustainability and Economic Assessment (Phase 3):

The strategies identified in Phase 2 were subjected to lifecycle evaluations.

Life Cycle Assessment (LCA):

An LCA was conducted to evaluate the environmental impact of proposed materials (e.g., Insulation, Windows, PV Panels) from cradle to grave without Module D. Module B6 was included to account for the net energy balance due to the respective renovation parameter. Key metrics included Global Warming Potential (GWP). (see Table 6)

Environmental Product Declarations (EPDs) for the insulation materials glass mineral wool (Knauf Insulation Sprl, 2023) and hemp fibre (Ekolution AB, 2025), window systems double-glazed (Aluminium Spanish Association (Asociación Española de Aluminio – AEA), 2020) and triple-glazed (Elitfönster AB, 2024), and photovoltaic modules (Jinko Solar Co., 2025) were obtained from manufacturer databases and used to calculate the LCA inventory data.

Table 6: Lifecycle Assessment (LCA) Data - Global Warming Potential (GWP)

Impact Category	insulation1_GMW GWP-Fossil /kg CO ₂ eq.	insulation2_Hemp GWP-Fossil /kg CO ₂ eq.	Window1 GWP-Fossil /kg CO ₂ eq.	Window 2 GWP-Fossil /kg CO ₂ eq.	PV Panel GWP-Fossil /kg CO ₂ eq.
A1 - A3	1.760	-2.090	117.000	61.600	0.342
A4	0.152	0.028	1.230	2.170	0.0243
A5	0.087	0.105	0.004	0.170	0.0015
C1-C4	0.038	0.001	1.280	4.444	0.0135

Life Cycle Cost (LCC) Analysis:

An economic model was built in Excel to calculate the Net Present Cost of each major renovation scenario. Inputs included initial investment costs, estimated operational energy savings, maintenance costs, and discounted interest rate. This analysis identified the most economically viable solutions over the building's lifespan and a defined study period of 50 years. (see Table 7)

Table 7:: Lifecycle Cost (LCC) Data - Material and Labour Cost Estimate

	GMW/ (SEK/m ²)	Hemp/(SEK/m ²)	Window1 /(SEK/m ²)	Window 2 /(SEK/m ²)	PV Panel/(SEK/ Watt peak)
Product Cost Supply Only	100	180	2500	3400	15
Labour Cost	175		1250		3.5

The present value of the total lifecycle cost is obtained using the standard LCC annuity and present-value formulas (Refer Equation 1) with a real discount rate of 3.14 percent over a 50-years analysis period, consistent with the 3 percent real discount rates commonly applied in Swedish and European building-energy cost-optimality studies. (“Cost-optimality guidelines EN annex,” 2025)

$$PV_i = C_i \frac{(1 + r_{di})^n - 1}{r_{di} \cdot (1 + r_{di})^n} \quad \text{Equation 1}$$

PV_i Present value of an expenditure occurring after N years [SEK]
 C_i Expenditure in today's value [SEK]
 r_{di} Discount interest rate [%]
 n Number of years [years]

Cost data for window systems, including materials and labour, were obtained from Elitfönster and Husexperter. Prices for insulation materials (Glass Mineral Wool and Hemp) were sourced from Beijer Byggmaterial and Bygghemma. Photovoltaic (PV) system cost assumptions were based on national Swedish survey reports. (Lindahl, 2023)

2.4 Result Integration (Phase 4):

The workflow for the environment and economic analysis was integrated into the parametric renovation design system. The outputs from combined simulations were carefully analysed to understand how different passive, active, and behavioural measures affect building performance. For each renovation scenario, key parameters were compiled and compared to see patterns and interactions across all cases.

The parametric workflow in Grasshopper was essential for this process, as it ensured all scenarios were evaluated simultaneously and made it easy to test additional parameters or performance targets if needed. To obtain a 3-dimensional relation, the results were normalised

and incorporated with equal weighting on each performance criterion. A synchronised trend Pareto front function was generated to prioritize an optimal set of renovation measures.

RESULTS AND DISCUSSION

The resulting parameter set showed differing outcomes when evaluated individually across the three performance criteria: energy use, LCC, and LCA. For the Building Energy Performance, a simple measure like PV Peak with (16.9 kWp) delivers the lowest operational energy demand (see Fig. 7). For the LCC category, the base case (no interventions) naturally shows the lowest cost (see Fig. 9). In the LCA category, individual product analysis shows that 300 mm thick Hemp insulation placed on the interior side shows the lowest embodied carbon emission (see Fig. 8). Whereas, when operational energy(B6) impact is included, PV Peak with (16.9 kWp) outperforms all individual measures. (see Fig. 10).

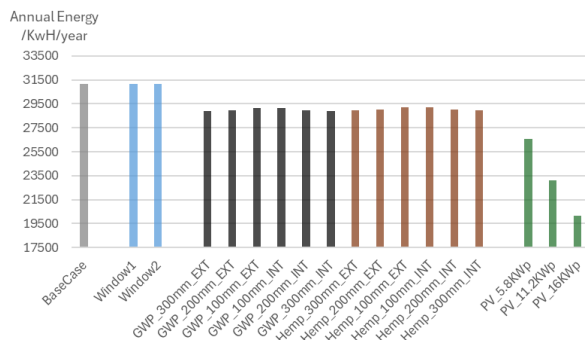


Fig. 7: Results of Energy for individual parameters.

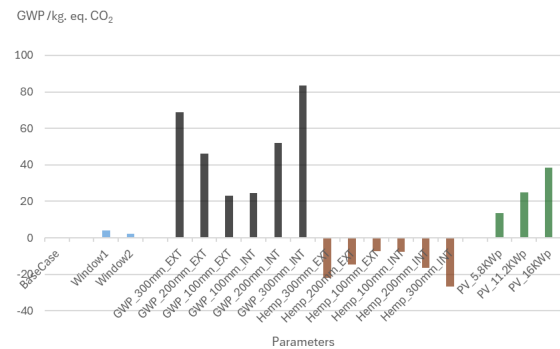


Fig. 8: Results of LCA (excluding B6) for individual parameters

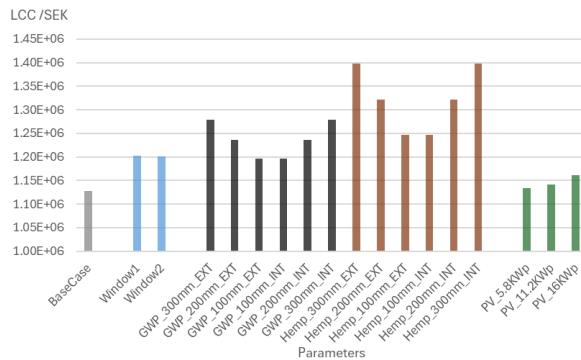


Fig. 9: Results of LCC for individual parameters

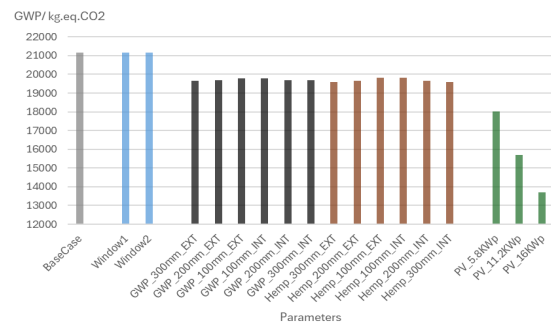


Fig. 10: Results of LCA (including B6) for individual parameters

When analysing the results for passive, active, and behavioural strategies, active strategies reduce significantly more energy compared to passive strategies, with energy reduction in both office and apartment exceeding 30%. For LCC criteria, a 5.8 kWpeak photovoltaic system represents the most cost-effective solution with only 0.5% additional cost. Although hemp insulation shows the lowest LCA as a product, an active system of a 16.9 kWpeak photovoltaic (inclusive of B6) reduces carbon emissions by approximately 35% compared to the base case.

Among the total 324 parametric sets simulated, 61 and 32 optimized data sets for the existing and proposed functions were analysed, respectively. When trade-offs between pairs, energy vs.

LCA, energy vs. LCC, LCC vs. LCA across both office and apartment scenarios were examined, the output suggested a direct relation between LCA and energy. The best performing criteria (as two-dimensional pareto front) for the trade-offs can be seen in Fig. 11, Fig. 12 and Fig. 13.

The blue markers represent the solutions for the office function, whereas the orange markers represent the solutions for the apartment function. The Pareto solutions for each function are indicated by square markers in the corresponding colour.

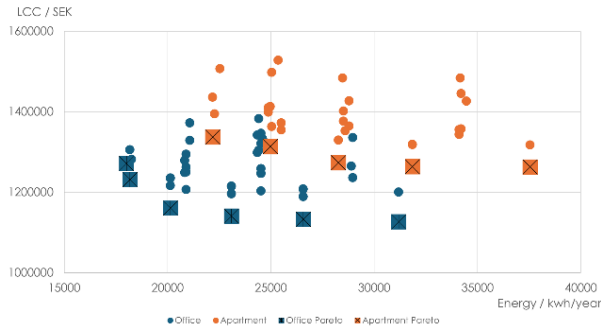


Fig. 11: Results of Energy vs LCC for both functions (office and apartment)

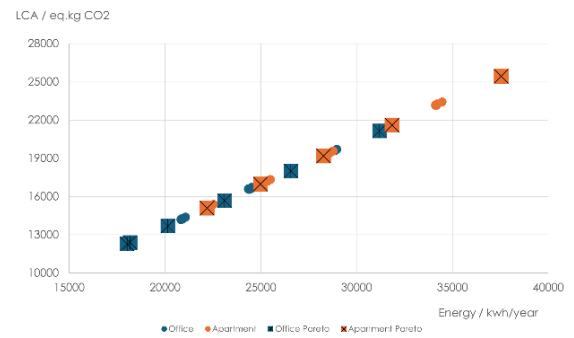


Fig. 12: Results of Energy vs LCA for both functions (office and apartment)

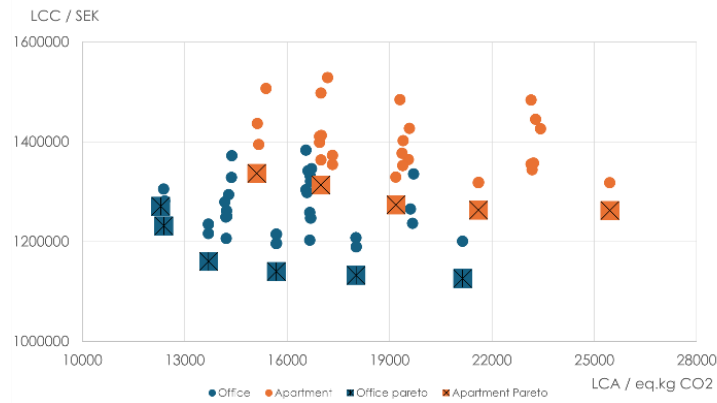


Fig. 13: Results of LCC vs LCA for both functions (office and apartment)

The non-dominated trends or the Pareto front trends from the 3-dimensional relation provided an improved set of renovation measures, balancing all the performance conditions. This set of renovation measures for each category, existing and proposed functions, is illustrated through the graphs (Fig. 14 & Fig. 15), where the Annual Energy is plotted on the Y-axis, the Life Cycle Cost on the X-axis, and the Environmental Impact is indicated by the size of the bubbles. Under each category, each combination is considered a case, further subcategorized into Excellent, Good, and Poor, based on their overall performance. The base case appears as the large red bubble in the upper-left corner, representing the existing building without window upgrades, no additional insulation, or PV. The excellent cases are labelled as per the respective performance ranking. These ranked best renovative measures for each building function can be seen in Table 8 for the Office function and Table 9 for the Apartment function, respectively. The energy saving with respect to the cost has been mentioned to finalize any of the 6 options for the office and any of the 7 sets of results for the apartment.

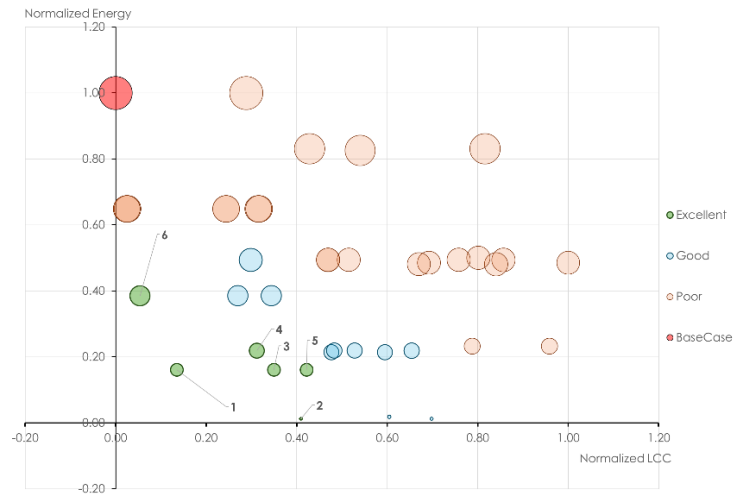


Fig. 14: Three-Dimensional Pareto front for optimized results for Office

Table 8: Optimal Renovation Measures for Office

Rank	Window Type	Insulation Type	Insulation Side	Insulation Thickness	PV Peak	Energy saving	Extra Cost
1	Existing	No Change	No Change	No Change	16.9KWp	35%	3%
2	Existing	GMW	Exterior	100mm	16.9KWp	42%	9%
3	Double	No Change	No Change	No Change	16.9KWp	35%	8%
4	Existing	GMW	Exterior	100mm	11.1KWp	33%	7%
5	Triple	No Change	No Change	No Change	16.9KWp	35%	10%
6	Existing	No Change	No Change	No Change	11.1KWp	26%	1%

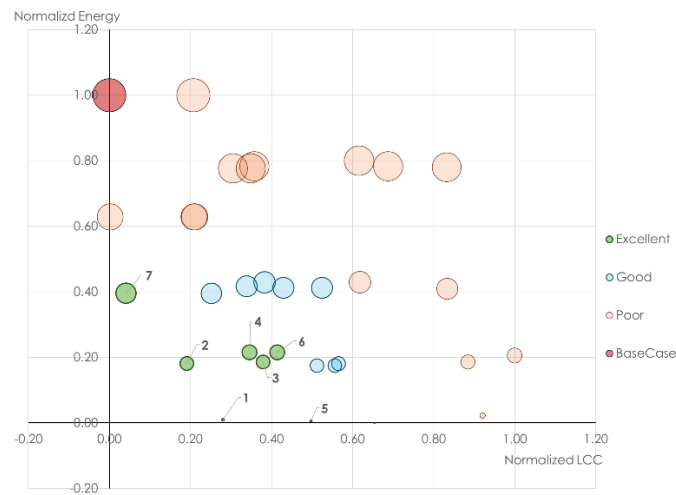


Fig. 15: Three-Dimensional Pareto front for optimized results for Apartment

Table 9: Optimal Renovation Measures for Apartment

Rank	Window Type	Insulation Type	Insulation Side	Insulation Thickness	PV Peak	Energy saving	Extra Cost
1	Existing	GMW	Interior	100mm	16.9KWp	41%	6%
2	Existing	GMW	Interior	100mm	11.1KWp	33%	4%
3	Existing	Hemp	Exterior	100mm	11.1KWp	33%	8%
4	Double	No Change	No Change	No Change	16.9KWp	32%	7%
5	Double	GMW	Exterior	100mm	16.9KWp	41%	10%
6	Triple	No Change	No Change	No Change	16.9KWp	32%	9%
7	Existing	No Change	No Change	No Change	11.1KWp	25%	1%

Discussion

Single-objective optimization exposes fundamental conflicts: cost minimization opposes energy and carbon reduction. The Apartment requires window upgrades for operational efficiency but not for embodied carbon minimization, whereas the Office maintains unchanged windows across all objectives. Both buildings converge on maximum PV (16.9 kWp), confirming active measures outperform passive upgrades.

For the Office, envelope improvements (upgraded windows or added insulation) increase cost and embodied carbon without lowering energy demand meaningfully. Internal heat gains from occupants, lighting, equipment dominate annual consumption, making PV investment far more effective than passive measures. The Apartment shows different behaviour. Window upgrades and reduced PV shift designs away from the Pareto front toward higher cost and energy use. The optimal balance combines modest passive measures (basic insulation, existing windows) with substantial PV. Window upgrades add cost without sufficient operational compensation, while reducing PV leaves the building energy dependent.

Across both buildings, premium glazing and insulation beyond 200 mm show diminishing returns, with embodied carbon and cost increasing while operational improvements remain minimal. This reflects retrofit reality: once the envelope reaches moderate efficiency, further passive upgrades become economically and environmentally inefficient. Large PV systems, by contrast, consistently improve performance across all objectives. Hemp insulation appears least on the best scenarios for either building. Despite lower embodied carbon, its moderate thermal efficiency and higher cost fail to outperform glass mineral wool.

Limitations

Several limitations should be considered when interpreting the results of this renovation assessment. The building model was simplified in several respects. The roof inclination angle was set to 44.5° for energy simulations rather than the actual 53°, although the correct angle was applied for PV generation calculations. The ventilation system was also simplified, with only the air-change rate adjusted while the overall system configuration remained unchanged. Additionally, occupancy schedules were based on estimated usage patterns rather than measured data, introducing the possibility of discrepancies between predicted and actual energy consumption. The optimization process using Octopus contains inherent limitations that may introduce gaps in the results.

The LCA assessment considered only the insulation layer instead of the entire wall system. Global Warming Potential, excluding other environmental impact categories, has been analyzed. Future electricity price fluctuations and economic conditions introduce uncertainty, and the discount rate of 3.14% applied may not adequately represent long-term economic variability.

CONCLUSIONS

The results show that active measures, mainly PV systems, deliver better outcomes than expensive passive upgrades for both buildings. The Office should go purely active: max PV, keep the windows as they are, and skip the extra insulation. The Apartment works better with a mix: add some insulation and pair it with a large PV system. Upgrading windows and adding thick insulation just costs more money and creates more embodied carbon without saving much energy in practice. What works depends on the building itself. If it's mostly heated by internal gains like the Office, PV is the way to go. If heating demand matters more, like in the Apartment, modest insulation alongside PV makes sense. In addition, the priority of the stakeholder is the most important key factor in deciding which performance criteria to consider most.

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